Employee Manager (Console + ADO.NET)

Да се направи конзолно приложение със следното меню:

```
1. List employees (paging)
2. Find employee by id
3. Search by name
4. Add employee
5. Update salary
6. Delete employee
0. Exit
```

Структура на проекта

```
Program.cs
Employee.cs
EmployeeRepository.cs
EmployeeService.cs
```

Model:

```
public class Employee
{
    public int EmployeeID { get; set; }
    public string FirstName { get; set; } = null!;
    public string LastName { get; set; } = null!;
    public decimal Salary { get; set; }
    public int DepartmentID { get; set; }

}
```

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Repository:

```
public class EmployeeRepository
{
 private readonly string connectionString;

 public EmployeeRepository(string cs)
 {
     connectionString = cs;
 }

 public List<Employee> GetAll(int count)
 {
     var result = new List<Employee>();

     using var con = new SqlConnection(connectionString);
     con.Open();
```

```
    string sql = @"
SELECT TOP (@count)
    EmployeeID, FirstName, LastName, Salary, DepartmentID
FROM Employees
ORDER BY EmployeeID";

    using var cmd = new SqlCommand(sql, con);
    cmd.Parameters.AddWithValue("@count", count);

    using var reader = cmd.ExecuteReader();

    while (reader.Read())
    {
        result.Add(new Employee
        {
            EmployeeID = reader.GetInt32(0),
            FirstName = reader.GetString(1),
            LastName = reader.GetString(2),
            Salary = reader.GetDecimal(3),
            DepartmentID = reader.GetInt32(4)
        });
    }

    return result;
}

public Employee? GetById(int id)
{
    using var con = new SqlConnection(connectionString);
    con.Open();

    string sql = @"SELECT EmployeeID, FirstName, LastName, Salary, DepartmentID
               FROM Employees WHERE EmployeeID = @id";

    using var cmd = new SqlCommand(sql, con);
    cmd.Parameters.AddWithValue("@id", id);

    using var reader = cmd.ExecuteReader();

    if (!reader.Read()) return null;

    return new Employee
    {
        EmployeeID = reader.GetInt32(0),
        FirstName = reader.GetString(1),
        LastName = reader.GetString(2),
        Salary = reader.GetDecimal(3),
        DepartmentID = reader.GetInt32(4)
    };
}

public void Add(Employee e)
{
    using var con = new SqlConnection(connectionString);
    con.Open();

    string sql = @"INSERT INTO Employees(FirstName, LastName, Salary, DepartmentID)
```

```
                VALUES (@fn, @ln, @sal, @dep)";

    using var cmd = new SqlCommand(sql, con);
    cmd.Parameters.AddWithValue("@fn", e.FirstName);
    cmd.Parameters.AddWithValue("@ln", e.LastName);
    cmd.Parameters.AddWithValue("@sal", e.Salary);
    cmd.Parameters.AddWithValue("@dep", e.DepartmentID);

    cmd.ExecuteNonQuery();
}

public void UpdateSalary(int id, decimal salary)
{
    using var con = new SqlConnection(connectionString);
    con.Open();

    string sql = @"UPDATE Employees SET Salary = @sal WHERE EmployeeID = @id";

    using var cmd = new SqlCommand(sql, con);
    cmd.Parameters.AddWithValue("@sal", salary);
    cmd.Parameters.AddWithValue("@id", id);

    cmd.ExecuteNonQuery();
}

public void Delete(int id)
{
    using var con = new SqlConnection(connectionString);
    con.Open();

    using var tx = con.BeginTransaction();

    try
    {
        string sql = "DELETE FROM Employees WHERE EmployeeID = @id";
        using var cmd = new SqlCommand(sql, con, tx);
        cmd.Parameters.AddWithValue("@id", id);
        cmd.ExecuteNonQuery();

        tx.Commit();
    }
    catch
    {
        tx.Rollback();
        throw;
    }
}

public List<Employee> Search(string text)
{
    var result = new List<Employee>();

    using var con = new SqlConnection(connectionString);
    con.Open();

    string sql = @"SELECT EmployeeID, FirstName, LastName, Salary, DepartmentID
                FROM Employees
                WHERE FirstName LIKE @txt OR LastName LIKE @txt";
```

```
    using var cmd = new SqlCommand(sql, con);
    cmd.Parameters.AddWithValue("@txt", "%" + text + "%");

    using var reader = cmd.ExecuteReader();

    while (reader.Read())
    {
        result.Add(new Employee
        {
            EmployeeID = reader.GetInt32(0),
            FirstName = reader.GetString(1),
            LastName = reader.GetString(2),
            Salary = reader.GetDecimal(3),
            DepartmentID = reader.GetInt32(4)
        });
    }

    return result;
 }
}
```

==========

Service:

```
public class EmployeeService
{
    private readonly EmployeeRepository repo;

    public EmployeeService(EmployeeRepository r)
    {
        repo = r;
    }

    public void IncreaseSalary(int id, decimal percent)
    {
        var emp = repo.GetById(id);
        if (emp == null) return;

        decimal newSalary = emp.Salary * (1 + percent / 100);
        repo.UpdateSalary(id, newSalary);
    }
}
```

Main:

```
static void Main(string[] args)
 {
    string cs =
"Server=(localdb)\\MSSQLLocalDB;Database=SoftUni;Trusted_Connection=True;";
    var repo = new EmployeeRepository(cs);
    var service = new EmployeeService(repo);

    while (true)
    {
```

```
        Console.Clear();
        Console.WriteLine(" 1.List\n 2.Find\n 3.Search\n 4.Add\n 5.Update Salary\n
6.Delete\n 0.Exit");
        var input = Console.ReadLine();

        if (input == "0") break;

        switch (input)
        {
            case "1":
                Console.Write("How many employees to show: ");
                int count = int.Parse(Console.ReadLine()!);

                var list = repo.GetAll(count);
                foreach (var el in list)
                    Console.WriteLine($"{el.EmployeeID} {el.FirstName}
{el.LastName} {el.Salary}");
                Console.ReadLine();
                break;

            case "2":
                Console.Write("ID: ");
                int id = int.Parse(Console.ReadLine()!);
                var emp = repo.GetById(id);
                if (emp == null) Console.WriteLine("Not found");
                else Console.WriteLine($"{emp.FirstName} {emp.LastName}
{emp.Salary}");
                Console.ReadLine();
                break;

            case "3":
                Console.Write("Search text: ");
                string txt = Console.ReadLine()!;
                var found = repo.Search(txt);
                foreach (var el in found)
                    Console.WriteLine($"{el.EmployeeID} {el.FirstName}
{el.LastName}");
                Console.ReadLine();
                break;

            case "4":
                var e = new Employee();
                Console.Write("First name: ");
                e.FirstName = Console.ReadLine()!;
                Console.Write("Last name: ");
                e.LastName = Console.ReadLine()!;
                Console.Write("Salary: ");
                e.Salary = decimal.Parse(Console.ReadLine()!);
                Console.Write("DepartmentId: ");
                e.DepartmentID = int.Parse(Console.ReadLine()!);
                repo.Add(e);
                Console.ReadLine();
                break;

            case "5":
                Console.Write("ID: ");
                int id2 = int.Parse(Console.ReadLine()!);
                Console.Write("Percent: ");
```

```
                decimal p = decimal.Parse(Console.ReadLine()!);
                service.IncreaseSalary(id2, p);
                Console.ReadLine();
                break;

            case "6":
                Console.Write("ID: ");
                int id3 = int.Parse(Console.ReadLine()!);
                repo.Delete(id3);
                Console.ReadLine();
                break;
        }
    }
}
```